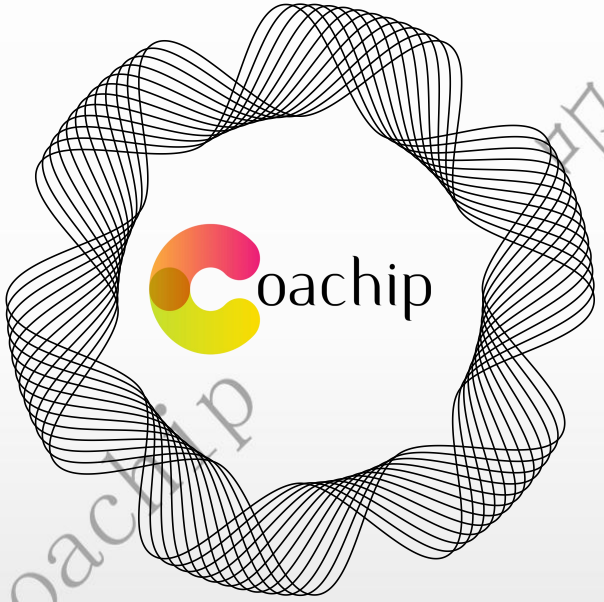


SV DPI

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课程内容及目标

课程内容：

- 了解什么是DPI
- 了解DPI的基本用法

目标：

- 能理解DPI
- 能通过DPI引入C的函数

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SV DPI 简介

DPI (Direct Programming Interface)

- An interface between SystemVerilog and a foreign programming language: C or C++
- Simple interface to C models
  - Allows SystemVerilog to call a C function just like any other native SystemVerilog function/task
  - Variables passed directly from/to C/C++
- Why DPI ?
  - SystemVerilog users have C/C++ indesigns and testbench that tehy want to reuse
  - DPI easily connectes C/C++ code to SV
- Support both function and tasks
- SystemVerilog can call C and C can call SV

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从C import task/function到SV

Import “DPI-C”

- SystemVerilog calling C/C++ task/function
- C code must have #include <svdpi.h>

```
// SystemVerilog code
program automatic test;
  import "DPI-C" context task c_test(input int addr);

  initial c_test(1000);

endprogram
```

```
// C code
#include <svdpi.h>

void c_test(int addr) {
   ...
}
```

```
> vcs -R -sverilog top.sv c_test.c
```

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从SV export task到C

Export “DPI-C”

- C calls SystemVerilog function and task
- C code must have #include <svdpi.h>

```
program automatic test;
  import "DPI-C" context task c_test(input int addr);
  initial c_test(1000);
  export "DPI-C" task apb_write;

  task apb_write(input int addr, data);
    ... @(posedge ready); ...
  endtask
endprogram
```

```
#include <svdpi.h>
extern void apb_write(int, int);

void c_test(int addr) {
  ...
  apb_write(addr, data);
  ...
}
```

```
> vcs -sverilog top.sv c_test.c
```

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SV DPI 例1

利用DPI 调用C code 来批量访问register

```
import "DPI" context task c_test(input int base_addr);
program automatic top;
    semaphore bus_sem = new(1);
    export "DPI" task apb_write;
    task apb_write(input int addr, data);
        bus_sem.get(1);
        #10 $display("VLOG : APB Write : Addr = %x, Data = %x  ", addr, data);
        bus_sem.put(1);
    endtask
```

```
initial begin
    fork
        c_test(32'h1000);
        c_test(32'h2000);
    join
end
endprogram
```

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SV DPI 例1

C调用APB write 驱动

```
#include <svdpi.h>
extern void abp_write(int, int);

void c_test(int base) {
    int addr, data;
    for(addr=base; addr<base+5; addr++) {
        data = addr*100;
        apb_write(addr, data);
        printf("C_TEST: APB_Write:  addr = 0x%8x, data = 0x%8x\n",addr, data);
    }
}
```

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SV DPI 例1

Makefile的改动：

我们不能再用vlogan来compile，所以之前的COMP和ELAB分步编译就不行了

必须用

```
> vcs -sverilog
```

```
1  TOP = top
2
3  DFILES = test.c test.sv
4
5  SEED = 1
6  VCOM  =  vcs -full64 -R -sverilog -debug_access+all -l elab.log -sim_res=1ps -Mupdate
7  RUN   =  $(TOP).simv -l run.log +ntb_random_seed=$(SEED) &
8
9  comp:
10         $(VCOM) $(DFILES)  -top $(TOP) -o $(TOP).simv
11
12 run:
13         ./$(RUN)
14
15 clean:
16         \rm -rf *.log
17         \rm -rf AN.DB
18         \rm -rf *simv*
19         \rm -rf DVEfiles
20         \rm -rf csrc
21         \rm -rf *vpd
22         \rm -rf ucli.key
```

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SV DPI 例 2

利用DPI 调用C code 间接调用PERL脚本

```
import "DPI-C" function int call_perl(string s);

program automatic top;
    int ret_val;
    string script;

    initial begin
        if(!$test$plusargs("script"))begin
            $display("NO +script switch found");
            $finish;
        end
        else begin
            $value$plusargs("script=%s",script);
            $display("SV: Running '%s'",script);
            ret_val = call_perl(script);
            $display("SV: Perl script return %0d",ret_val);
        end
    end
endprogram : top
```

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SV DPI 例 2

利用DPI 调用C code 间接调用PERL脚本

```
#include <svdpi.h>
#include <stdlib.h>
#include <wait.h>

int call_perl(const char* command){
    int result = system(command);
    return WEXITSTATUS(result);
}
```

DPI重要的一点是可以调用外部其他语言，最简单的办法就是调用system()任务，如果需要命令的返回值，使用linux的system()函数和WEXITSTATUS任务（需要#include <wait.h>）。

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SV DPI 例2

PERL脚本

```
1 #!/usr/bin/perl
2
3 print "Perl:Hello World! \n";
4 exit (3)
```

需要在make file的 RUN option添加：`+script="perl hello.pl" &`

这样才能调用perl命令。

所以，如果要调用tcl、python等其他语言，只要在+script=这里改成其他语言即可。

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SV DPI 例2

Makefile脚本update

```
1  TOP = top
2
3  DFILES = test.c test.sv
4
5  SEED = 1
6  VCOM  =  vcs -full64 -R -sverilog -debug_access+all -l elab.log -sim_res=1ps -Mupdate
7  RUN   =  $(TOP).simv -l run.log +ntb_random_seed=$(SEED) +script="perl hello.pl" &
8
9  comp:
10         $(VCOM) $(DFILES)  -top $(TOP) -o $(TOP).simv
11
12 run:
13         ./$(RUN)
14
15 clean:
16         \rm -rf *.log
17         \rm -rf AN.DB
18         \rm -rf *simv*
19         \rm -rf DVEfiles
20         \rm -rf csrc
21         \rm -rf *vpd
22         \rm -rf ucli.key
```

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练习

练习前面讲的例1和例2

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Q&A

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